

Incident Response Action Plan

Incident Response, Containment, Eradication, Recovery, and Lessons Learned

Analyst: Suraj Somkuwar

Project: Hands-On PCAP Analysis and Threat Detection Using Generative AI

Dataset: Combined PCAP-derived network traffic dataset

Total records analysed: 3,588

Primary investigation host: 10.10.0.33

1. Incident Overview

The network traffic investigation was based on a single PCAP file provided during the initial stage of the project. In Milestone 1, the PCAP file was converted into structured CSV data. In Milestone 2, the resulting CSV dataset was divided into three segments to support AI-assisted traffic analysis and investigation. In Milestone 3, these three analysed CSV segments were consolidated into a single Combined_Traffic_Data.csv dataset for combined traffic analysis.

The combined dataset contains 3,588 traffic records, including 3,549 IP-bearing records and 39 ARP/non-IP records. The consolidated dataset was subsequently used for statistical analysis, detection-rule development, network traffic visualization, comparative analysis, and timeline analysis.

Milestones 2 and 3 identified several communication patterns that are relevant for further security investigation. The most significant finding was the concentration of communication around internal host 10.10.0.33, which was identified as the primary internal/private host involved in the observed public-unicast communication.

The combined analysis identified:

- 3,525 true internal ↔ public-unicast communication records
- 1,785 internal-to-public records
- 1,740 public-to-internal records
- 1,725 bidirectional records involving 10.10.0.33 and 18.184.26.60
- 804 TCP/53480 records
- 112 DNS records
- 4 statistical frame-size outliers above the IQR-based upper threshold
- Repeated and bursty communication patterns
- A high-volume one-minute interval containing 2,097 records

These findings are considered investigation-relevant indicators rather than confirmed evidence of compromise.

The available packet metadata does not independently establish confirmed malware, command-and-control communication, data exfiltration, DNS tunnelling, attacker identity, or malicious intent.

Therefore, the incident response strategy developed in this milestone follows a precautionary and evidence-based approach. The objective is to contain potentially risky activity while preserving

evidence and conducting additional endpoint, DNS, threat-intelligence, and security-log validation before making a final compromise determination.

2. Incident Classification and Current Security Posture

2.1 Current Classification

Classification: Investigation required / investigation-relevant

The observed traffic warrants additional investigation and precautionary containment because communication is highly concentrated around a primary internal host and several external endpoints.

Compromise status: No confirmed compromise

The classification is deliberately different from a confirmed malware incident.

The PCAP-derived metadata provides information about source and destination relationships, communication frequency, ports, decoded protocols, frame lengths, timestamps, and traffic direction.

It does not, by itself, provide sufficient evidence to conclusively determine whether malware executed, whether an attacker controlled the system, whether a particular communication was command-and-control, whether data was actually exfiltrated, whether DNS tunnelling occurred, whether the observed traffic was unauthorized, or whether the external endpoints were malicious.

Consequently, the response plan should initially treat the activity as investigation-relevant network behaviour requiring validation.

3. Key Investigation Findings Supporting the Response Plan

3.1 Primary Internal Host

The primary internal/private host identified during the investigation was 10.10.0.33. This host is classified as an RFC1918 private IPv4 address. The investigation identified it as the primary internal host participating in the observed public-unicast communication. Because the host is central to the observed traffic relationships, it should receive the highest priority for endpoint and network investigation.

3.2 Internal ↔ Public Communication

The combined traffic analysis identified 3,549 IP-bearing records. These records can be separated into true internal-to-public communication and special local network traffic:

Traffic Category	Records	% of IP-Bearing Records
Internal to Public	1,785	50.30%
Public to Internal	1,740	49.03%
True Internal ↔ Public	3,525	99.32%
Local Broadcast (10.10.0.255)	16	0.45%
Multicast (239.255.255.250)	8	0.23%

Traffic Category	Records	% of IP-Bearing Records
Total IP-Bearing Records	3,549	100%

The 3,525 true internal ↔ public records represent communication between the primary internal host and public unicast IP addresses. They account for 99.32% of all IP-bearing records.

The remaining 24 IP-bearing records consist of 16 local broadcast records involving 10.10.0.255 and 8 multicast records involving 239.255.255.250. These were excluded from the internal ↔ public calculation because broadcast and multicast traffic are special local/network-discovery communication patterns rather than ordinary public-unicast internet communication.

This demonstrates that the captured IP-bearing traffic is overwhelmingly associated with internal-to-public or public-to-internal communication. The concentration is investigation-relevant, but communication volume alone does not establish malicious activity.

4. Dominant Communication Relationship

The strongest observed communication relationship was 10.10.0.33 and 18.184.26.60. The relationship contained 877 records from 10.10.0.33 to 18.184.26.60, and 848 records from 18.184.26.60 to 10.10.0.33, for 1,725 records combined. The 1,725 records represent approximately 48.08% of the complete 3,588-record dataset. This relationship therefore represents the highest-priority communication relationship identified during the investigation.

Recommended Response

The external endpoint should be investigated using approved threat-intelligence information, historical network baseline, approved communication/allow lists, security logs, application context, and endpoint telemetry. The response team should determine whether the communication is expected and business-related before permanently blocking the destination.

5. Important Port Observations

Several destination ports were identified as significant in the combined dataset.

Destination Port	Records	% of All Records
TCP/80	1,221	34.03%
TCP/53480	804	22.41%
TCP/443	506	14.10%
TCP/53483	181	5.04%
UDP/53	58	1.62%

5.1 TCP/53480

TCP/53480 is particularly important because 804 records were observed using this destination port. The analysis established that all 804 of the 804 records were observed in the relationship 18.184.26.60:80 to 10.10.0.33:53480.

Within these 804 records, the protocol composition was 803 records classified as eth:ethertype:ip:tcp and 1 record classified as eth:ethertype:ip:tcp:http:media. The additional http:media protocol layer in one record indicates that Wireshark/TShark identified HTTP media content within that particular TCP packet; it does not represent a different source/destination relationship.

This makes the traffic worthy of session reconstruction and endpoint-level investigation. The port number itself should not be treated as proof of malicious activity, because high-numbered ports can occur in legitimate network sessions.

Recommended action: Investigate which process or application was associated with the communication, whether the communication was expected, whether the session corresponds to legitimate web activity, whether the communication appears in endpoint/network logs, and whether the relationship existed before the captured period.

6. DNS Investigation

The combined dataset contained 112 DNS records, representing 3.12% of all records. The principal DNS relationship identified was 58 records from 10.10.0.33 to 8.8.8.8 and 54 records from 8.8.8.8 to 10.10.0.33, for a total of $58 + 54 = 112$ DNS records.

The frame-size investigation also identified four statistical outliers associated with DNS traffic: 52,160 bytes, 53,529 bytes, 54,032 bytes, and 58,337 bytes. The IQR-based upper threshold was 3,695 bytes. These observations require contextual investigation. However, the project findings explicitly do not establish that these records represent DNS tunnelling or DNS exfiltration.

Recommended DNS investigation: Examine DNS query logs, requested domains, query frequency, response characteristics, endpoint processes generating DNS requests, historical DNS behaviour, whether the DNS destinations are approved, and whether the observed large frames correspond to legitimate captured traffic characteristics. The purpose is to determine whether the DNS activity is normal, unusual, or potentially unauthorized.

7. Protocol Profile

The combined dataset was classified using mutually exclusive protocol categories.

Protocol	Records	Percentage
TCP (Remaining)	3,093	86.20%
TLS	201	5.60%
HTTP	119	3.32%
DNS	112	3.12%
NBNS	16	0.45%
SSDP	8	0.22%
ARP / Non-IP	39	1.09%
Total	3,588	100%

The protocol distribution shows that TCP was the dominant traffic category. HTTP and TLS represented the principal web-oriented traffic categories, while DNS represented a smaller portion of the total dataset. Protocol distribution alone does not establish malicious activity.

8. Temporal and Traffic-Volume Findings

The timeline analysis examined the distribution of the 3,588 records across one-minute intervals. The validated timeline contains 11 intervals. The capture began at 01:58:06.017228 +0530 and ended at 02:08:16.417023 +0530, for a total capture duration of approximately 10 minutes 10.40 seconds (10.1733 minutes).

The largest one-minute interval was 01:59 to 02:00, with 2,097 records. The remaining intervals were:

Interval	Records
01:58 to 01:59	696
01:59 to 02:00	2,097
02:00 to 02:01	54
02:01 to 02:02	82
02:02 to 02:03	113
02:03 to 02:04	68
02:04 to 02:05	104
02:05 to 02:06	81
02:06 to 02:07	103
02:07 to 02:08	150
02:08 to 02:09 (partial)	40

The final interval is partial because the capture ended at approximately 02:08:16.417023 +0530. The timeline demonstrates a substantial traffic burst during 01:59 to 02:00. However, a traffic-volume spike by itself does not establish malicious behaviour.

9. Beaconing Assessment

The timestamp analysis showed repeated and bursty communication. However, no stable periodic interval was established. Beaconing is therefore not confirmed.

This distinction is important for incident response. The response team should not classify the host as beaconing solely because repeated communication was observed. If endpoint investigation later identifies a process communicating at regular intervals, that additional evidence can be correlated with the network observations.

10. Frame-Size Investigation

The combined frame-size analysis identified a minimum of 42 bytes, a first quartile of 60 bytes, a median of 66 bytes, a mean of 672.06 bytes, a third quartile of 1,514 bytes, and an interquartile range of 1,454 bytes. The resulting upper IQR threshold was 3,695 bytes, against a maximum observed frame length of 58,337 bytes.

Four records exceeded the calculated upper threshold. The four exact outlier frame lengths were 52,160 bytes, 53,529 bytes, 54,032 bytes, and 58,337 bytes. These four records represented approximately 0.11% of all records. The four statistical outliers were associated with DNS traffic in the investigation.

Response interpretation: These records should be validated against the actual network/application context. The existence of unusually large frame-length values does not independently establish data exfiltration, DNS tunnelling, malware, command-and-control activity, or compromise. The appropriate action is contextual validation rather than immediate classification as malicious traffic.

11. Immediate Containment Actions

The following actions should be considered immediately if the investigation is being handled as a live security incident.

11.1 Isolate the Primary Internal Host

Priority: High

Temporarily isolate 10.10.0.33 from normal network communication while maintaining the system for forensic investigation. The isolation should be performed in a manner that preserves relevant evidence.

The purpose is precautionary containment because this host is associated with the dominant external communication relationship. Isolation should not be documented as proof that the system is compromised.

11.2 Preserve Network and Endpoint Evidence

Before destructive remediation actions are performed, preserve the original PCAP evidence, the three original segment datasets, the combined traffic dataset, analysis scripts, detection rules, investigation reports, visualization outputs, timeline analysis, and relevant logs. Evidence should be retained so that subsequent investigators can reproduce and validate the findings.

Where available, also collect and preserve running-process information, network connection information, endpoint security alerts, relevant application logs, DNS logs, security-event logs, system logs, and authentication logs. These sources can provide evidence that packet metadata alone cannot establish.

12. External Communication Validation

The investigation identified several external/public endpoints, including 18.184.26.60, 204.79.197.200, 88.221.0.167, 204.11.56.48, 34.102.136.180, and 8.8.8.8 for DNS activity. These addresses should not automatically be classified as malicious.

The response team should determine whether the destination is approved, whether it belongs to a legitimate service, whether the communication is expected for the affected host, whether the destination appears in threat-intelligence sources, and whether the communication is consistent with historical behaviour. If investigation confirms unauthorized or malicious communication, appropriate network restrictions can then be applied.

13. Containment Strategy

Containment should operate at both the host and network levels.

Host-level containment: For 10.10.0.33: isolate the system, preserve evidence, identify active processes, identify network connections, review endpoint-security alerts, and determine which applications generated the observed traffic.

Network-level containment: Where appropriate and supported by additional validation: restrict unauthorized external destinations, apply temporary firewall controls, restrict unnecessary outbound communication, review DNS communication, and monitor the relevant ports and destinations. Network controls should be based on validated evidence rather than IP address or port number alone.

14. Eradication Strategy

Eradication actions should only begin after evidence preservation and investigation. Because the PCAP evidence does not independently confirm malware, eradication should be conditional upon endpoint investigation findings.

Malware removal: If endpoint investigation identifies malicious software: remove the confirmed malicious component using approved security procedures, check for associated persistence mechanisms, and verify that the malicious process is no longer active.

System reimaging: If system integrity cannot be confidently established, reimage the affected workstation using a trusted system image. This provides a stronger recovery assurance than attempting to clean an uncertain system.

Credential protection: If investigation establishes credential exposure: reset affected credentials, revoke potentially compromised sessions or tokens where applicable, and review authentication activity. Credential resets should not be claimed as necessary solely because network traffic was observed.

Vulnerability remediation: If the endpoint investigation identifies an exploited or vulnerable application: apply the relevant security patches, update affected software, remove unnecessary vulnerable services, and confirm remediation before returning the system to normal operation.

15. Recovery Strategy

The recommended recovery lifecycle is:

ISOLATE -> INVESTIGATE -> REMEDIATE -> VALIDATE -> RECONNECT -> MONITOR

Stage 1, Isolate: Prevent potentially unauthorized communication while evidence is preserved.

Stage 2, Investigate: Correlate PCAP observations, endpoint telemetry, DNS logs, security logs, threat intelligence, and historical network behaviour.

Stage 3, Remediate: If compromise is confirmed: remove malware, remove persistence, patch vulnerabilities, reset affected credentials, and reimage where necessary.

Stage 4, Validate: Before reconnection: verify endpoint security controls, confirm expected processes, verify network connections, verify DNS behaviour, check for recurrence of suspicious communication, and compare behaviour against a known-good baseline.

Stage 5, Reconnect: Return the host to normal network operation only after validation criteria have been satisfied.

Stage 6, Monitor: Maintain heightened monitoring for at least 30 days. The monitoring should specifically look for recurrence of the investigation-relevant patterns identified during the PCAP analysis.

16. Recovery Validation Criteria

The system should not be considered fully recovered until:

- No confirmed malicious process remains
- Required remediation has been completed
- Security controls are operational
- Expected network communication has been established
- Unexpected external communication is investigated
- DNS behaviour is understood
- Relevant logs are being collected
- No recurrence of previously unexplained communication has been identified
- The system can be compared against an established normal baseline

17. Post-Incident Monitoring

The following indicators should receive additional monitoring during the recovery period.

Network relationships: Monitor communication involving 10.10.0.33 and the previously observed external destinations.

High-frequency communication: Monitor for recurrence of unusually concentrated communication relationships.

TCP/53480: Monitor for unexpected recurrence of the high-volume TCP/53480 pattern.

DNS: Monitor DNS volume, unusual query behaviour, unexpected domains, and abnormal response characteristics.

Traffic-volume spikes: Monitor for traffic bursts similar to the 01:59 to 02:00 interval, while remembering that traffic volume alone is not proof of malicious activity.

Frame-size anomalies: Monitor unusually large frame lengths and correlate them with application and protocol context.

Repeated communication: Monitor for regular periodic communication that could provide additional evidence for or against beaconing.

18. Lessons Learned

18.1 Data Preparation Is a Critical Security Step

The project demonstrated that reliable security analysis depends on correctly structured and validated traffic data. The three PCAP-derived datasets were consolidated into a combined dataset containing 3,588 records, enabling cross-segment analysis.

18.2 AI Can Accelerate Security Analysis

AI-assisted analysis helped identify and interpret dominant communication relationships, internal and external IP classifications, protocol distributions, port concentrations, frame-size patterns, traffic-volume patterns, and investigation priorities. This reduced the effort required to identify patterns within the combined traffic dataset.

18.3 AI Results Must Be Independently Verified

The project did not treat AI output as final evidence. The analytical results were checked against the underlying dataset and statistical calculations. This is particularly important because AI can identify a potentially interesting pattern without having sufficient evidence to determine its security meaning. LLMs are very good at spotting unusual trends, but they aren't always smart enough to know if a trend is an actual security threat or just a harmless coincidence.

18.4 Investigation-Relevant Does Not Mean Confirmed Malicious

One of the most important lessons from the project is the distinction between observed, investigation-relevant, and confirmed. The traffic contained several observations worthy of investigation. However, the available packet metadata did not independently establish malicious activity. This principle prevents analysts from over-classifying normal or unexplained network behaviour as an attack.

18.5 Visualization Improves Investigation

The project demonstrated how visual representations can make complex network traffic easier to understand. The completed visualizations include network traffic flow, protocol distribution, destination port usage, top communicating hosts, external communication, routine versus investigation-relevant traffic, frame-size analysis, detection-rule mapping, the overall security investigation dashboard, and timeline analysis. These visualizations provide different perspectives of the same underlying traffic dataset.

19. AI-Assisted Investigation and Verification

AI or LLM was used throughout the project as an analytical assistant. The workflow was:

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PCAP Data -> Structured CSV -> Segment Analysis -> Combined Dataset -> AI Analysis -> Statistical Validation -> Detection Rules -> Visualization -> Analyst Interpretation
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AI-assisted analysis was used to identify patterns and suggest areas requiring further investigation. Human/analyst verification was then used to validate record counts, IP relationships, protocol

classification, port observations, frame-size statistics, timeline counts, traffic direction, and security interpretation.

The final incident-response recommendations therefore rely on verified observations rather than unverified AI conclusions.

20. Final Incident Response Assessment

Based on the complete investigation performed across the project:

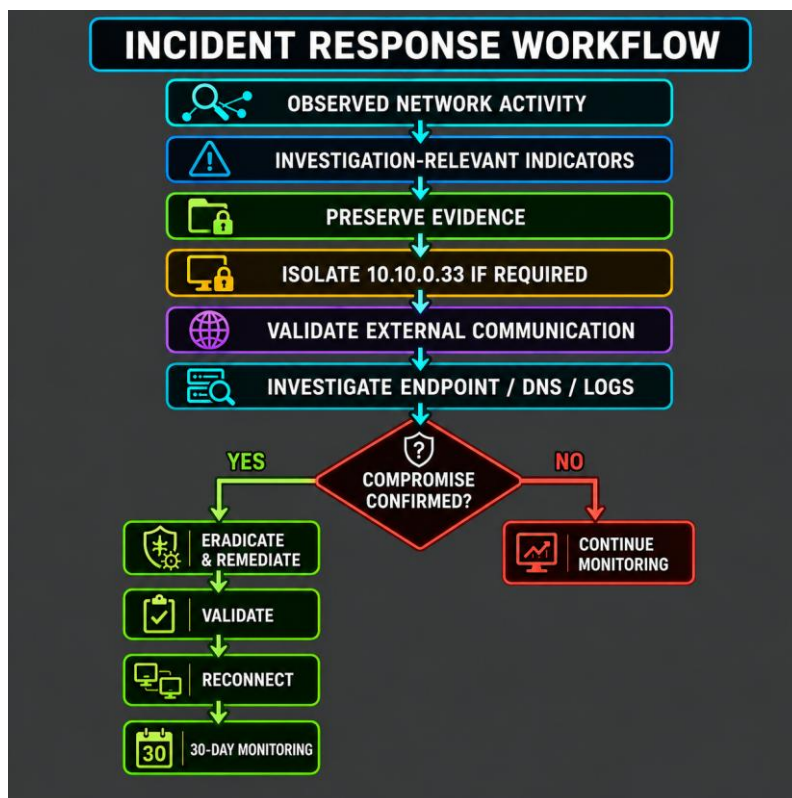
Item	Value
Current status	Investigation required
Primary host	10.10.0.33
Dominant external endpoint	18.184.26.60
Strongest observed relationship	10.10.0.33 ↔ 18.184.26.60, 1,725 records
Total internal ↔ public communication	3,525 records
TCP/53480	804 records
DNS	112 records
Statistical frame-size outliers	4 records
Beaconing	Not confirmed
Confirmed malware	Not established from available packet metadata
Confirmed C2	Not established from available packet metadata
Confirmed data exfiltration	Not established from available packet metadata

Final Response Position

The observed network activity contains investigation-relevant communication patterns that justify precautionary containment and additional validation. However, the available packet metadata does not independently establish confirmed malicious activity. Further endpoint, DNS, threat-intelligence, security-log, and baseline analysis is required before making a final compromise determination.

21. Recommended Incident Response Workflow

The complete recommended response can be summarized and visualized as follows.



Visual: Recommended Incident Response Workflow

This workflow ensures that the response is proportional to the available evidence.

22. Final Conclusion

The combined PCAP investigation identified a highly concentrated pattern of communication involving internal host 10.10.0.33 and several public external endpoints. The most significant relationship was between 10.10.0.33 and 18.184.26.60, accounting for 1,725 bidirectional records. Overall, 3,525 records represented true internal-to-public or public-to-internal communication.

Additional investigation-relevant observations included 804 TCP/53480 records, 112 DNS records, four statistical frame-size outliers, and a substantial traffic-volume burst of 2,097 records during the 01:59 to 02:00 interval. These observations provide sufficient justification for additional security investigation and precautionary containment.

However, the analysis also demonstrates an important limitation of packet metadata: the observed traffic cannot independently establish malware, command-and-control activity, data exfiltration, DNS tunnelling, attacker identity, or malicious intent. Therefore, the appropriate incident-response position is: investigation required, with no confirmed compromise established from available packet metadata alone.

The recommended response is to preserve evidence, prioritize investigation of 10.10.0.33, validate its external communication, perform endpoint and DNS investigation, use threat intelligence and security logs for correlation, and apply containment, eradication, and recovery measures according

to the evidence obtained. This approach ensures that the response remains technically accurate, evidence-based, and proportionate to the findings of the PCAP investigation.

23. Referenced Evidence and Visuals

The following visuals, produced during Milestone 3, support the findings summarized in this incident response action plan.

Figure 1: Overall Security Investigation Dashboard



Figure 2: External Communication Analysis

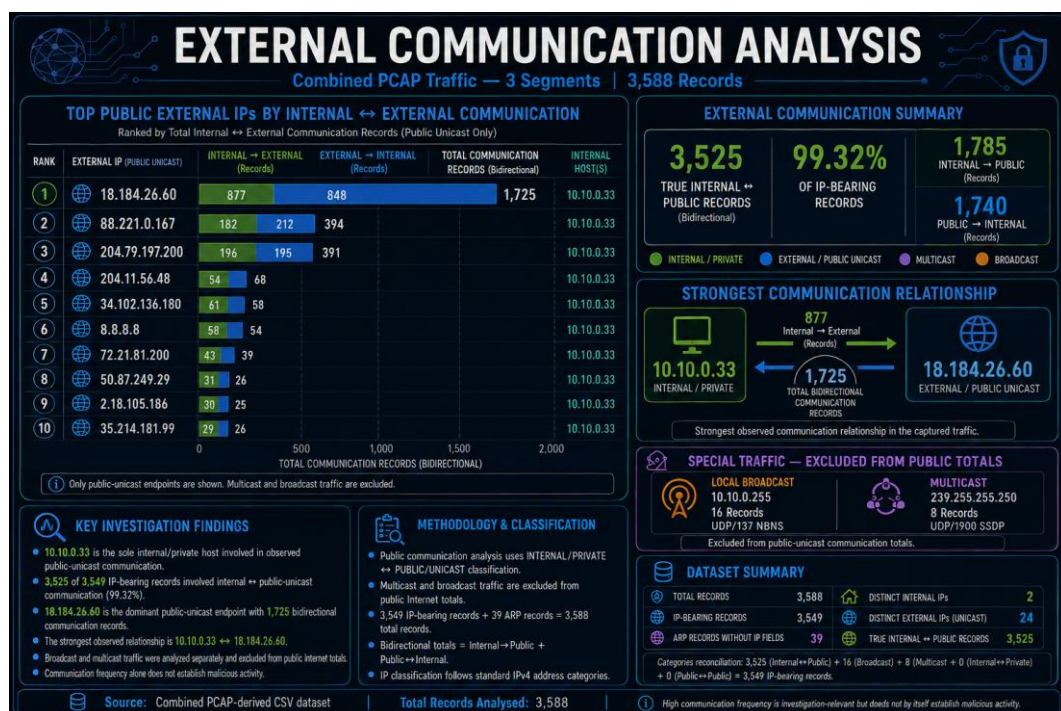


Figure 3: Detection Rule Mapping

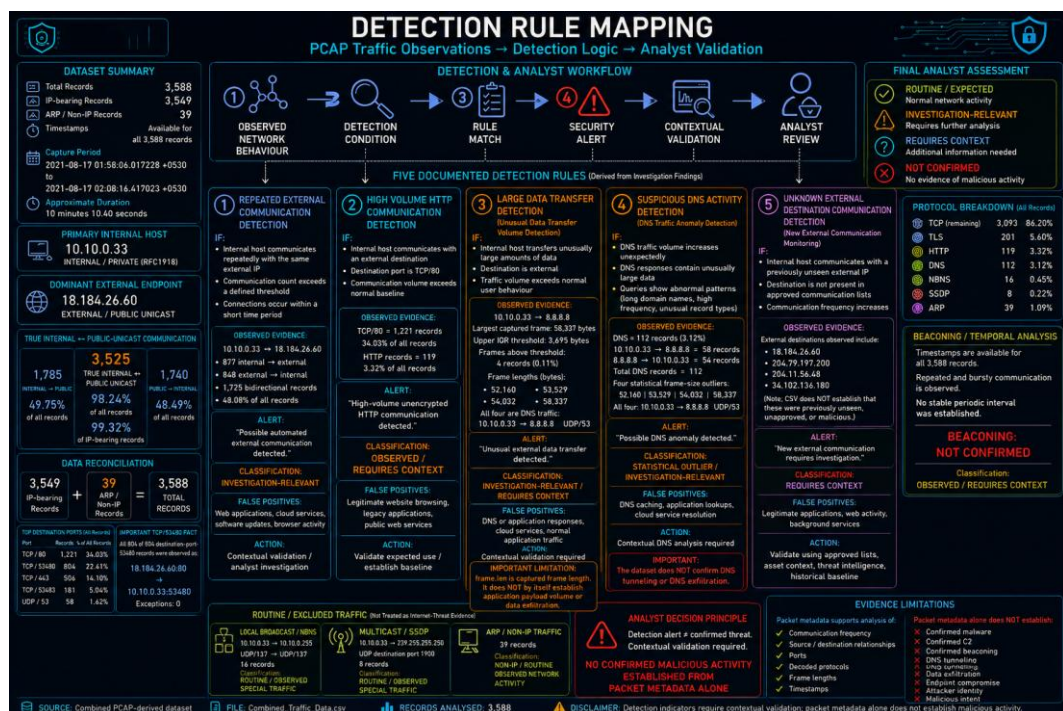


Figure 4: Network Traffic Timeline Analysis

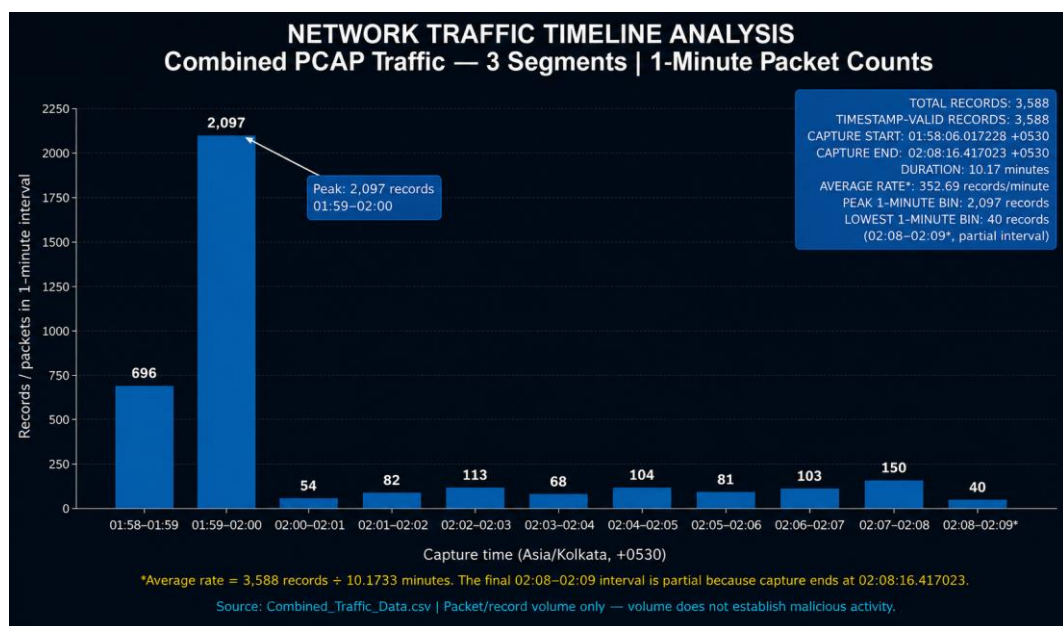


Figure 5: Protocol Distribution Analysis

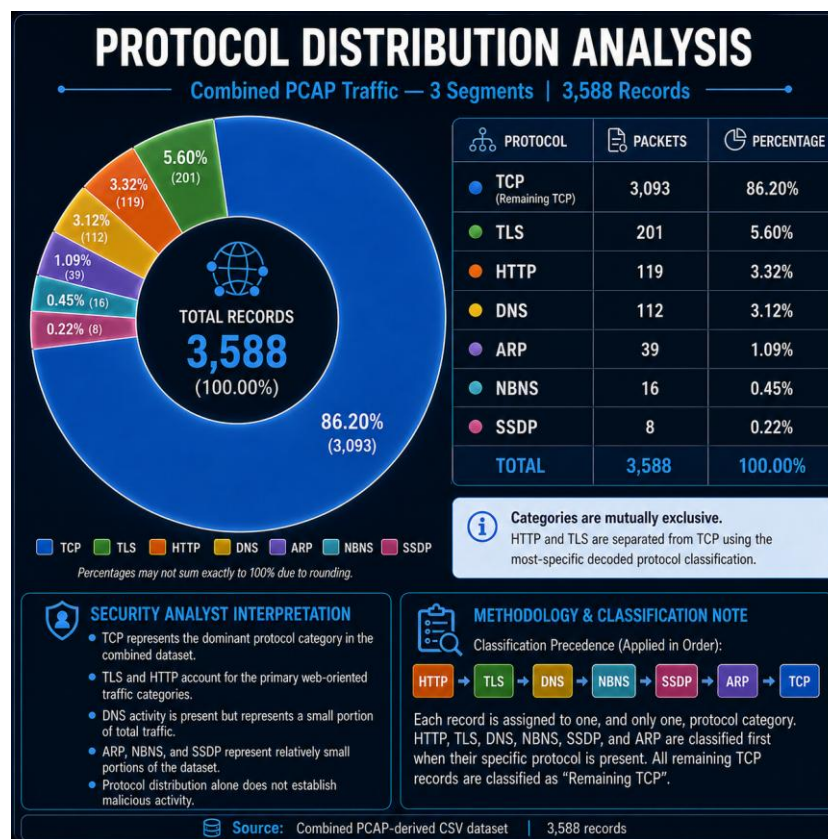


Figure 6: Frame Size Analysis

